

MICHAEL DOMM

U.S. Citizen

✉ mrdomm@uwaterloo.ca

TECHNICAL SKILLS

Microfabrication Tools; E-beam and UV Lithography, Plasma-Enhanced CVD, E-beam PVD, and Wet Benches.
Programming; MATLAB, Python, SolidWorks, LaTeX, LabVIEW, Lumerical, MEEP, and S4.
Tools; Oscilloscopes, Soldering Tools, Lasers, Mills, Lathes, MIG Welders.

EDUCATION

Institute for Quantum Computing (IQC)

Sept. 2025 - Sept. 2027

Master of Science (M.Sc.) in Physics

University of Waterloo, Waterloo, ON

- Researcher, Quantum Photonic Devices Lab (Dr. Michael Reimer) — working on quantum dot entangled photon sources and single-photon detectors, with experience in cleanroom fabrication and device characterization
- Extracurricular activities; IQC Quantum Startup Accelerator Network Lead, on Board of Directors for Graduate Student Association (GSA), representative for IQC-GSA, Graduate Committee Members on Student Appeals.

University of Waterloo

Completion Apr. 2024

Bachelor of Science (B.S.) in Honours Physics, Co-operative Education Program.

Waterloo, ON

- Extracurricular activities; Varsity Athlete Rowing +3 years, Student Representative, Volunteer Outers Club 2 years.

RESEARCH EXPERIENCE

Dr. Strickland Lab

Jan. – May. 2024

Research Intern, Department of Physics at University of Waterloo

Waterloo, ON

- Co-supervised by 2018 Nobel Laureate Dr. Donna Strickland, continued previous Harvard research project on nonlinear optical materials: performed data analysis of optical experiments and built transfer-matrix simulations of multilayer thin films to model transmission spectra.

Harvard University, Capasso Lab

May. 2023 – Jan. 2024

Research Intern, Department of Applied Physics

Cambridge, MA

- Worked in the research group of Federico Capasso. Specialized in photonics research, particularly in wave front engineering by metasurfaces and quantum-cascade lasers (QCLs).
- Pushing the limit of the number of holograms that a single metasurface can produce at the image plane.
- Investigated and tested a nonlinear optical material to work with a guided mode resonance to actively select desired wavelengths from a broad band light source.
- Conducted Finite-Difference Time-Domain (FDTD) simulations on 90,000 nano pillars using Lumerical and MATLAB to design metasurfaces, achieving targeted wavefront manipulation.
- Built and automated optical setups for metasurface characterizations and testing. From which data was collected for multiple upcoming academic publications.
- Attended Harvard Nano Fabrication seminars and Audited Harvard class: Electronic and Photonic Devices.

WORK EXPERIENCE

P&P Optica (PPO)

Sept. – Dec. 2021 & May. – Aug. 2022

Optics Technician

Waterloo, ON

- Reduced assembly and alignment time for a near-infrared spectrometer 80% through innovative mechanical designs.
- Designed, performed, and documented experiments to test new lighting profiles and optical components for spectrometer.
- Assisted with the production and alignment of the spectrometer on the company's smart imaging systems.
- Led the development and integration of an innovative lighting system to significantly enhancing spectrometer data quality for AI model accuracy and performance.
- Deployed as an onsite supervisor to ensure PPO's standards for operation of the quality assurance system and AI models were met for clients in Texas.

Teledyne DALSA

Jan. – Apr. 2021

2D Image Sensor Technician

Waterloo, ON

- Set up test platforms with various instruments and creating Python code for various quality tests.
- Programmed and tested a software correction algorithm in MATLAB to correct lens distortion.
- Executed test plans with a team to identify a wide variety of capabilities of a new camera built by the R&D team.
- Independently engineered and constructed a versatile test bench, enabling comprehensive quality assessments of diverse camera models ensuring high product standards prior to customer delivery.